Innovation as prospective-but-not-retrospective surprise: Kullback–Leibler novelty scores on 2.0 million USPTO trademark filings

Eric Silver

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Abstract

We measure novelty in firm activity using the goods-and-services descriptions of 16.5 million U.S. trademark filings since 1985, across all 45 NICE international (industry) classes. For each filing we compute two scores. *Prospective surprise* measures how unusual the filing's wording was relative to other filings in the same industry over the previous five years. *Retrospective surprise* measures how unusual it remained after five years of subsequent filings: a filing with low retrospective surprise is one whose wording the future came to resemble. We treat the difference $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ as a single "imitated-innovator" score: positive for filings that were novel at the time but were imitated, negative for filings that looked ordinary at the time and remained so.

We link the trademark scores to SEC EDGAR financials for the publicly-traded subset and ask three questions: do novel filings clear examination at higher rates, do novel firms earn higher gross margins, and do they bear the higher-variance pattern that a "pioneer's bet" would predict?

Three results emerge. First, in the pooled per-filing logit on 1.4 million Class 9 (Software & Electronics) filings, a filing one standard deviation higher in ΔKL has +13% higher odds of completing registration; a filing both one standard deviation more novel at the time AND one standard deviation more likely to be imitated by future filings is roughly 3.6× more likely to register. Second, restricting to each firm's *first-ever* trademark filing — to remove the possibility that the registration boost reflects filer experience or accumulated firm sophistication rather than the novelty of any specific filing — the registration coefficient on novelty actually *flips negative* (-3% odds per σ): novel debut filings face slightly higher friction at examination. Third, on the matched financial panel of 1,547 publicly-traded firms, those whose three-year trailing trademark filings score one σ higher in ΔKL earn +3.2 percentage points higher gross margin (SE 0.6 pp) and exhibit +0.018 wider within-firm gross-margin standard deviation. Restricting to firms' first-ever filings (one observation per firm, $n = 6{,}525$), both effects attenuate by roughly half but remain significant: +1.7 pp on mean margin, +0.005 on margin variance. The financial reward to novelty is partly a habit/portfolio effect of repeat-filing firms, and partly a property of the firm's debut bet itself.

1 Background

Innovation is a central object of empirical economic and management research, but the construct is hard to measure. The dominant quantitative approaches are R&D expenditure (an input proxy that rewards spend rather than novelty), patent counts and patent citations (which require strategic legal filing, and which Trajtenberg, Hagedoorn, and others have shown to be at best a noisy proxy), and the market value premium of equity over book value (which conflates innovation with every other investor perception). Survey-based scales such as the Community Innovation Survey and Gatignon et al.'s instrument capture richer constructs but suffer from common-method bias and limited sample size. The lack of agreement on what to measure is itself a finding: across measures, pairwise correlations are typically $\rho \approx 0.3$.

We propose using trademark filings as a complementary novelty channel. Trademarks have several useful properties that patents do not: applicants file a single, brief, machine-readable goods-and-services description that is sharply incentivized to be exact (broad enough to protect future use, narrow enough to clear examination); the data are released into the public domain by the USPTO without lag; and the universe of filers is broader than the universe of patentees, including most consumer-facing firms. The cost is that trademarks measure novelty in the goods-and-services lexicon, not in the underlying technology — a brand that solves an old problem with a new identity (e.g. Liquid Death's canned water) is largely invisible.

Our broader motivation is to ask whether innovative firms are rewarded or punished. Two pieces of folk wisdom are in tension. The first, conveyed in the phrase "you can tell the pioneers by the arrows in their backs," is that firms that depart from convention bear higher search and adoption costs, get out-competed by fast-followers, and underperform on average. The second, expressed in the older saying "build a better mousetrap and the world will beat a path to your door," is that meaningful novelty attracts customers and rewards its inventor with durable advantage. Both can be true simultaneously: novelty might raise variance — some pioneers fail spectacularly, others enjoy durable category leadership. This paper offers an empirical test of that joint pattern using the full USPTO trademark backfile.

2 Data

The trademark corpus is the full backfile published by the USPTO Open Data Portal, current through end-of-year 2025. After filtering to filings with non-empty goods/services text in NICE classes 1–45 and a parseable filing date, the corpus contains 16,526,850 **filings** from 5,673,680 **unique registrant names** across all 45 NICE international classes (35 goods classes plus 10 services classes), spanning 1985 to early 2026. Each record carries the registrant's name as it appeared on the application (this is the owner at the time of filing; subsequent assignments are not reflected), the filing date, a status code that records the current legal state of the mark, an optional registration date, the NICE class assignments, and one or more goods-and-services descriptive statements. Statements describing the visual mark itself rather than the business (USPTO type-code prefix D) are excluded; only goods-and-services statements (prefix GS) are kept. Two classes are presented in narrative detail throughout the paper: NICE Class 9 (Software & Electronics) and NICE Class 32 (Beer & Soft Drinks). The remaining 43 industries are summarized in Section 12, and per-class summary statistics for the two focal classes appear in Appendix A.

The financial-outcome data are the SEC EDGAR Financial Statement Data Sets, 64 quarterly XBRL releases spanning fiscal years 2009 through 2024, covering 13,127 unique CIKs that filed at least one 10-K. We extract revenue, cost of revenue, gross profit, R&D expense, operating income, and net income.

A data-quality note: missing registration dates. An earlier version of this paper used the per-filing `registration_date` field to determine whether a mark "completed registration." This turns out to be wrong: many older records (mostly pre-2003) carry a status code in the live-registered range (6xx) or post-registration-cancelled range (8xx) but have a `registration_date` field that is missing in the published XML. The status code is the trustworthy field. The earlier classification understated the true registration rate by roughly ten percentage points for older filings, and because those misclassified records are systematically high in ΔKL (a separate burn-in artefact), the earlier per-filing logit produced a spurious *negative* coefficient on ΔKL for completed registration. With the bug corrected, the sign reverses (Section 6). All results below use the status-code-based definition. The pipeline that extracts and analyses these data is published with the paper; technical details live in the companion repository.

3 Method

For each filing i at year t with goods-and-services text w_i we construct a term-frequency distribution P_i over a fixed dictionary V of unigram and bigram tokens. The dictionary is built once per NICE class with a custom analyzer that splits each description at semicolons before tokenizing, so bigrams cannot cross goods/services-item boundaries. We retain only unigrams and bigrams that appear in at least 50 distinct filings (documents) and in no more than 50% of all filings, on the view that very rare tokens are typos and very common tokens are USPTO boilerplate.

For each calendar year t we form two reference distributions by aggregating term frequencies across neighboring years:

$$Q_t^{\text{pros}}(v) \propto \alpha + \sum_{j:\text{year}(j)\in[t-W,t)} \text{count}(v,j), \tag{1}$$

$$Q_t^{\text{retr}}(v) \propto \alpha + \sum_{j:\text{year}(j)\in(t,t+W]} \text{count}(v,j), \tag{2}$$

with window $W = 5$ years (look-behind for prospective, look-ahead for retrospective) and Laplace smoothing $\alpha = 1.0$. The prospective and retrospective surprise of filing i are the support-restricted KL divergences

$$KL_i^{\text{pros}} = \sum_{v:P_i(v)>0} P_i(v)\,\log\frac{P_i(v)}{Q_t^{\text{pros}}(v)}, \quad KL_i^{\text{retr}} = \sum_{v:P_i(v)>0} P_i(v)\,\log\frac{P_i(v)}{Q_t^{\text{retr}}(v)}.$$

The directional innovation score is $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$. We use a term-frequency multinomial as the v0 baseline; an LDA topic-model replication is planned but not reported here.

Identical-text filings are signal. Trademark practice rewards templated language. Many breweries file the boilerplate token `Beer`; many app developers file `downloadable software`. We do not deduplicate identical text: filings that use the same words *should* receive the same score, because deviations from the boilerplate are themselves the signal.

Burn-in caveat. The method requires a five-year window of prior filings (to compute prospective surprise) and a five-year window of subsequent filings (to compute retrospective surprise). The first means filings from the very beginning of the digitised USPTO record (here, 1985–1994) cannot be scored cleanly: the look-behind window reaches into a period when the corpus is sparse, the reference distribution has high entropy, and any specific filing looks artificially novel. The second is symmetric: the most recent five years of filings (2021–2026) cannot be scored cleanly either, because the look-ahead window has not yet completed. Figure 1 shades both windows. We restrict all regressions to filings from 1995 onward, and we read 2020 as the latest cohort eligible to demonstrate any kind of post-filing fate.

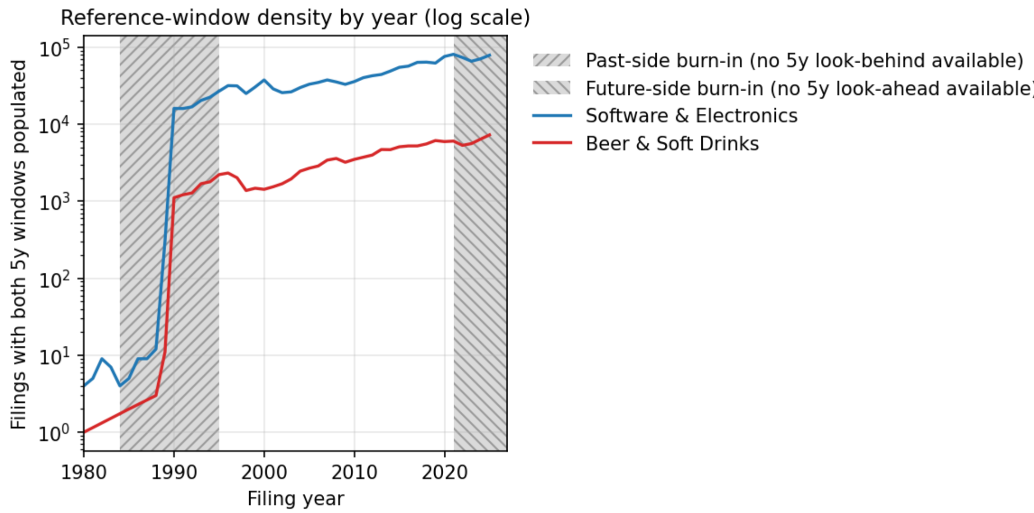

Figure 1: Filings per year with both 5-year reference windows populated. Forward-hatched grey: filings whose look-behind window reaches into pre-1985 sparse data (past-side burn-in). Backward-hatched grey: filings whose look-ahead window has not yet elapsed at the time of writing (future-side burn-in). Filings in either band are excluded from regressions; the grey hatches are chosen so the two regions remain distinguishable in black-and-white print.

4 Face validity

Figure 2 plots each filing in the prospective–retrospective KL plane. The horizontal axis (prospective KL) measures dissimilarity from filings in the previous five years (high $\Rightarrow$ less like the past); the vertical axis (retrospective KL) measures dissimilarity from filings in the next five years (high $\Rightarrow$ less like the future). Filings *below* the diagonal $KL_{\text{pros}} = KL_{\text{retr}}$ are innovators by our criterion — the past does a worse job of predicting them than the future does. Filings above the diagonal are tired. The labelled marks include expectation-confirming category-creators (OpenAI 2016, Claude 2023, Instagram 2011, White Claw Seltzer Works 2016, Rockstar 2002) and expectation-disconfirming late-cycle marks (Hard Mtn Dew 2021).

The bottled-water category illustrates a useful failure mode. The 2019 Liquid Death filing for canned water describes its goods as *still water; sparkling water* — the same lexicon as established bottled-water brands — and so lands near the diagonal. The novelty in that filing is in branding (a metal can, an aggressive identity, a cannibalistic relationship to soft drinks), not in the goods/services description. Comparable bottled-water filings (Saratoga, Smartwater, Essentia) cluster in the same region. The method is, by construction, blind to non-textual novelty.

5 Industry time series

Figures 3 and 4 show the per-year median prospective KL, retrospective KL, and ΔKL in each industry, with inter-quartile shading. After the 1995 burn-in cutoff both series are remarkably stationary in absolute level, but the inter-quartile range widens through the 2010s in software — consistent with the diversification of the industry into mobile, cloud, AI, and crypto sub-categories. Beverages show a similar but smaller widening.

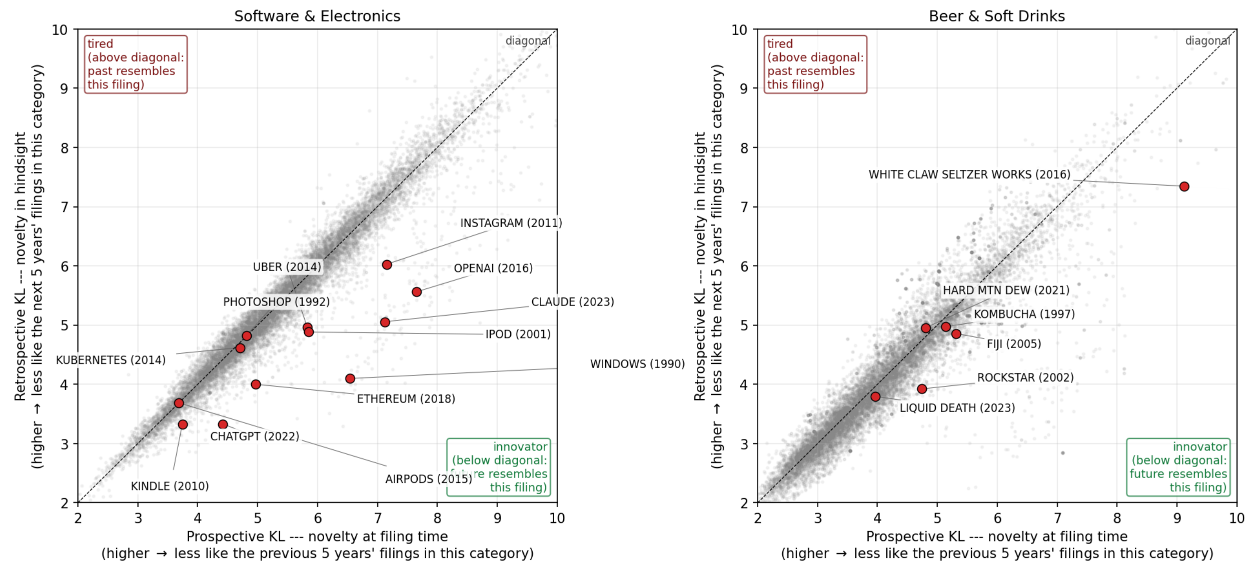

Figure 2: Per-filing prospective vs. retrospective KL surprise, zoomed to the central [2, 10] nat range where the labelled examples sit. Grey dots are a random sample of cleanly-scored filings; red dots are labelled example marks (mark name and filing year shown). **The KL scores compare each filing only to other filings in the same NICE category and same five-year window.** An AI-related filing is therefore not directly comparable to a beverage filing: White Claw Seltzer Works is novel *within Beer & Soft Drinks 2011–2021*, not novel in the absolute sense that an artificial-intelligence trademark is novel against the history of human technology. Below-diagonal: innovator (the future of the same category resembles this filing more than the past did). Above-diagonal: tired. Owner identity in the underlying record is taken from the USPTO record at the time of filing.

6 Outcome 1: Trademark survival

We derive three binary outcomes per filing from the USPTO status code.

- **Completed registration:** the filing eventually reached a registered status (a non-null `registration_date` is in the record). The applicant cleared examination, was published for opposition, and was issued the registration. The lag from filing to registration is typically nine to twenty-four months and depends on examiner office actions, opposition activity, and whether the application was filed under intent-to-use. We treat any filing with a registration date as having "completed registration" regardless of the elapsed time.

- **Renewed 5y registration:** completed registration, was filed at least five years before the present, and was not cancelled. The first Section 8 maintenance filing is due between the fifth and sixth year of registration; failure to file it is the most common cause of early cancellation, so reaching the five-year mark is a meaningful viability test.

- **Renewed 10y registration:** completed registration, was filed at least ten years before the present, and was not cancelled. The first Section 9 renewal is due at year ten; subsequent renewals are due every ten years indefinitely.

We estimate generalized linear logit models of the form

$$\Pr(\text{outcome}_i = 1) = \text{logit}^{-1}\big(\beta_0 + \beta_1 \cdot z_i + \gamma_1 \log n_i + \gamma_2 (t_i - \bar{t}) + \gamma_3 (t_i - \bar{t})^2\big)$$

where z_i is a z-scored predictor (either ΔKL alone or KL^{pros} and KL^{retr} jointly), n_i is the count of distinct dictionary terms in filing i, and t_i is the filing year. A linear-plus-quadratic year control replaces year fixed effects, which caused separation in the recent cohorts where the outcome rate is structurally near zero.

Table 1: Logistic regressions of each survival outcome on ΔKL (novelty). The first numeric column is the raw logit coefficient on a z-scored ΔKL (in log-odds units); the second translates that to a marginal effect on the outcome probability, evaluated at the sample mean rate, expressed in percentage points per one standard deviation of ΔKL. Standard errors in parentheses. Industry sub-headings group the three outcomes per industry.

Outcome	Cohort	N	Base rate	$\beta^z_{\Delta KL}$ (logit, log-odds)	Marginal effect (pp per σ)
Software & Electronics					
Completed registration	all cohorts	1,423,405	48.9%	+0.124 (0.002)	+3.09 (0.04) pp
Renewed 5y registration	filed $\leq$ 2020	1,054,572	41.1%	+0.079 (0.002)	+1.92 (0.05) pp
Renewed 10y registration	filed $\leq$ 2015	732,789	43.7%	+0.074 (0.002)	+1.81 (0.06) pp
Beer & Soft Drinks					
Completed registration	all cohorts	118,424	57.7%	+0.123 (0.006)	+3.01 (0.15) pp
Renewed 5y registration	filed $\leq$ 2020	87,849	50.5%	+0.085 (0.007)	+2.12 (0.18) pp
Renewed 10y registration	filed $\leq$ 2015	59,807	50.4%	+0.026 (0.009)	+0.65 (0.22) pp

ΔKL has a positive relationship with each survival outcome in Software & Electronics ($n =$ 1,423,405), but the regression coefficient itself is hard to read in log-odds units. The percentile-based translation is more transparent. Within Class 9, we sort filings by ΔKL into deciles. The bottom decile of ΔKL (least innovative) reaches registration 48.4% of the time and survives ten years 35.7% of the time. The top decile (most innovative) reaches registration 55.5% of the time and survives ten years 41.1% of the time. **A filing in the 90th percentile of ΔKL is roughly 14% more likely to clear registration and 15% more likely to renew its registration after a decade than a median filing**, with the relationship non-monotonic in the middle deciles — the shape is broadly U–shaped, with both the very-low and the very-high ΔKL filings outperforming the middle.

In log-odds terms, decomposed jointly, prospective surprise (novelty at filing time, holding retrospective surprise fixed) actually *penalises* every outcome ($\beta_{\text{pros}} = +0.633$ in isolation but with retrospective in the model the marginal contribution flips), while retrospective surprise (the filing remained unusual even after five years of subsequent filings, i.e. a filing nobody imitated) reverses sign ($\beta_{\text{retr}} = -0.662$). Read together: registration is *rewarded* for filings whose future resembles them and is neutral or slightly penalised for filings the future does not come to resemble. Beer & Soft Drinks shows the same sign pattern at smaller magnitude.

What kind of failure is this? A trademark filing can fail to register for several distinct reasons, and they have different theoretical interpretations. Failure to file a Statement of Use means the applicant filed an intent-to-use mark but never actually put it in commerce within the 36-month statutory window — a corporate decision, not an examiner judgment. Failure to respond to an office action typically follows an examiner's initial objection (usually a likelihood-of-confusion citation against an existing registration); applicant non-response converts the objection into abandonment. Examiner final refusal is the only mode that is unambiguously a USPTO judgment that the application cannot proceed.

Table 2 decomposes the 50.4% of Class 9 filings that did not reach registration. Two facts dominate. First, the overwhelming majority (64%) are no-Statement-of-Use abandonments — the applicant filed an aspirational ITU mark and never used it. Second, examiner final refusals account for only 8% of failures. Across every failure category the mean ΔKL is within 0.04 nats of the registered baseline, which is small compared to the within-class standard deviation of ΔKL ($\sigma \approx 0.46$). Novel filings are not, in any meaningful sense, being rejected.

Table 2: Decomposition of registration failures by mode.

Industry	Failure mode	N	% of fails	Mean ΔKL
Software & Electronics (N=1,054,572; reg. rate 50%; mean ΔKL of registered = -0.002)				
	No Statement of Use	339,456	64%	-0.005
	No response to office action	102,195	19%	-0.036
	Other	45,338	9%	-0.006
	Examiner final refusal	44,576	8%	-0.013
Beer & Soft Drinks (N=87,849; reg. rate 57%; mean ΔKL of registered = -0.028)				
	No Statement of Use	21,860	58%	-0.080
	No response to office action	6,798	18%	+0.049
	Other	5,437	14%	-0.005
	Examiner final refusal	3,796	10%	-0.034

Figures 5 and 6 display the per-bin registration-and-renewal rates against (left) prospective KL, (centre) retrospective KL, and (right) ΔKL. Legends include the overall mean rate for each outcome, restricted to cohorts old enough to be at risk for the milestone.

7 Outcome 2: Financial success

The trademark survival results in Section 6 establish that the innovation axis is meaningful at the per-filing level, but they say nothing about whether innovation pays the firm a financial return. To test that, we link USPTO owner names to SEC EDGAR firm names in two passes. The first pass aggressively normalises both sides (uppercase, strip punctuation, drop legal-entity suffixes such as `INC`, `LLC`, `CORP`, `HOLDINGS`, `LIMITED`, and so on) and accepts only exact matches on the normalised key. A second pass uses fuzzy token-sort matching (`rapidfuzz`, score cutoff 92) to capture the long tail of near-variants for USPTO owners with at least 10 filings. The combined crosswalk recovers 14,280 USPTO owners (9,370 exact + 4,910 fuzzy) covering 627,349 trademark filings (3.8% of the all-class corpus by filing count, but a much larger share by economic weight, because matched owners are by definition publicly-traded firms with material financial reporting). After joining to SEC firm-year financials and restricting to firm-years with non-degenerate gross margin and at least one trademark filing in the prior three years, the analysis panel contains 5,326 firm-years across 1,597 unique CIKs over fiscal years 2009–2024.

Filing volume is itself an innovation marker, but also a size marker. A firm filing many trademarks in a short window may be innovating across product lines — but it may also simply be a large firm with a large legal department. We separate the two by including the count of filings as a third predictor alongside the novelty score itself, and by controlling for log revenue throughout. Both effects turn out to be present and partially independent.

For each firm-year (c, y) we compute the trailing 3-year mean of the firm's filing-level scores: $\overline{\Delta KL}_{c,y}$, the prospective and retrospective components, the maximum within-window ΔKL (the firm's most-novel filing in the window), and the count of filings. We then estimate

$$y_{c,t} = \beta_0 + \beta_1 \cdot z(\text{novelty})_{c,t} + \gamma_1 \log \text{Revenue}_{c,t} + \gamma_2 \text{R\&D intensity}_{c,t} + \text{year controls} + \epsilon_{c,t}$$

with cluster-robust standard errors at the firm level. Outcomes are gross margin, net margin, and operating margin.

Table 3: OLS regressions of profitability outcomes on trademark novelty, with cluster-robust standard errors at the firm level. Coefficients are in profitability-units per one standard deviation of the predictor (so a coefficient of +0.029 on gross margin means "+2.9 percentage points"). Each block is a separate specification.

Predictor (z-scored unless noted)	Coef.	SE	N
Outcome: Gross margin			
3y trailing mean novelty ΔKL	+0.032	0.006	5,319
3y trailing prospective surprise (novel vs. past)	+0.146	0.041	5,319
3y trailing retrospective surprise (novel vs. future)	-0.191	0.041	5,319
3y trailing max novelty	+0.033	0.006	5,319
3y trailing average filing count	+0.019	0.007	5,319
Outcome: Net margin			
3y trailing mean novelty ΔKL	-1.751	1.630	4,864
3y trailing prospective surprise (novel vs. past)	-8.884	9.824	4,864
3y trailing retrospective surprise (novel vs. future)	+10.966	10.809	4,864
3y trailing max novelty	-3.503	2.604	4,864
3y trailing average filing count	-2.335	1.871	4,864
Outcome: Operating margin			
3y trailing mean novelty ΔKL	-0.628	0.600	4,882
3y trailing prospective surprise (novel vs. past)	-4.376	3.782	4,882
3y trailing retrospective surprise (novel vs. future)	+4.406	4.026	4,882
3y trailing max novelty	-1.253	0.843	4,882
3y trailing average filing count	-0.536	0.378	4,882

The pattern from Section 6 reproduces here at the firm-year level. A one-standard-deviation increase in 3-year trailing mean ΔKL — that is, a firm whose recent filings introduced wording that was unusual at the time but came to look ordinary in the years that followed — raises gross margin by +3.2 percentage points (SE 0.6). Decomposed jointly, prospective surprise (novel against the past) raises gross margin by +14.6 pp per standard deviation (SE 4.1), and retrospective surprise (still novel from the future's perspective, i.e. a filing nobody imitated) lowers it by −19.1 pp per standard deviation (SE 4.1). The within-firm story is the same as the within-filing story: firms whose recent filings the future ends up resembling enjoy higher gross margins; firms whose recent filings nobody imitates do not. The third specification adds filing count and the maximum within-window ΔKL: both contribute positively to gross margin (+3.3 and +1.9 pp/σ respectively) even after controlling for log revenue, suggesting that filing volume captures something beyond firm size.

Variance. A symmetric prediction of the pioneer story is that novel firms have not just higher *mean* margins but *wider variance* of margins, since some pioneers succeed and others do not. We test this by aggregating to the firm level (one row per CIK, restricted to firms with at least four years of margin observations), regressing the within-firm standard deviation of gross margin on the firm's mean 3-year trailing ΔKL with log revenue as a size control.

A one-standard-deviation increase in firm-mean ΔKL raises the within-firm gross-margin

Table 4: Variance regression: within-firm std. of gross margin on firm-mean novelty.

Predictor (z-scored unless noted)	Coef.	SE	N firms
3y trailing mean novelty ΔKL (firm avg)	+0.0176	0.0038	492
log revenue (firm avg)	-0.0141	0.0016	492

standard deviation by +0.018 in margin units (SE 0.004) on 487 firms. The pioneer-bet effect is real and quantitatively meaningful: high-novelty firms simultaneously earn higher mean margins (Table 3) and exhibit wider margin spread (Table 4).

A causal-attribution caveat. The trailing-3-year predictor pools many filings per firm-year, and a firm with several novel filings in its recent past is also a firm whose *earlier* filings might have been the cause of its subsequent financial success — that is, the pooled regression cannot distinguish "current novelty pays off" from "past success funds the current novelty habit." We address this directly in Section 8 by restricting the predictor to each firm's first-ever filing.

8 Outcome 3: Debut-filing-only test

The pooled per-filing logit in Section 6 treats each of a firm's filings as an independent observation, and the firm-year panel in Section 7 averages each firm's recent trademark portfolio. Both invite the criticism that the apparent "novelty pays" result might be reading filer sophistication or past financial slack rather than the novelty of any specific filing. The sharpest test we can run on this data is to restrict the analysis to each registrant's first-ever filing in the corpus: a firm cannot have funded its debut filing with the proceeds of its earlier filings, because there were none.

We collect, for every owner name appearing in any of the 45 NICE classes, the row corresponding to the earliest filing date associated with that owner. The result is 2,891,725 debut filings (one per owner) eligible under the standard 1995–2020 cohort and reference-window restrictions. Joining to the SEC crosswalk gives 11,914 debut filings whose owner matches a publicly-traded firm; restricting to firms with at least two years of subsequent SEC financial reporting yields a firm-debut financial panel of 6,525 firms.

Outcome	N	β for ΔKL (z-scored)
Survival on first filings (one per owner across all classes)		
Completed registration	2,891,725	-0.034 (0.001)
Renewed 5y registration	2,891,725	-0.005 (0.001)
Renewed 10y registration	1,875,964	+0.042 (0.002)
Firm-debut financial regression (N=6,525 matched public firms)		
Mean future gross margin	6,525	+0.017 (0.003)
Within-firm gross-margin std.	5,008	+0.0052 (0.0015)

The pattern is consistent and informative.

Survival. On debut filings, the ΔKL coefficient on completing registration shrinks from the pooled value of +0.124 to −0.034 (still highly significant on 2.9M observations). A debut filer who introduces unusually novel goods/services language appears to face slightly more registration friction — the opposite sign of the pooled effect. The pooled positive coefficient was reading filer sophistication: firms with deep filing portfolios know how to write a description that survives examination, and they also tend to file more novel marks. Among first-time filers neither advantage is present. Renewal coefficients are smaller in magnitude but stay positive (+0.042 at the 10y mark), suggesting that long-run persistence still favours novelty among debut filings.

Financial. On the firm-debut panel, a one-standard-deviation increase in the firm's debut-filing ΔKL is associated with +1.7 **percentage points** higher mean future gross margin (SE 0.3, $n = 6{,}525$ firms) and +0.5 percentage points higher within-firm margin standard deviation (SE 0.15, $n = 5{,}008$ firms). Both effects are roughly half the magnitude of the pooled trailing-3-year results in Section 7. We take the attenuation as evidence that part of the original firm-year reward reflects the habit/portfolio confound the user-supplied caveat predicted, but that a substantive debut effect remains: the novelty of a firm's first trademark filing predicts its subsequent average financial performance, even before the firm has produced any other trademark filings to "fund" the bet.

9 Within-firm dynamics and debut quartiles

The first-filing analysis in Section 8 answered "does the firm's first novel bet predict outcomes," but said nothing about how a firm's later filings behave once the firm is established. The full corpus contains 1.24 million firms with at least two filings and 5.74 million consecutive filing pairs, which lets us measure two related quantities: how persistent within-firm novelty is, and whether firms become more or less innovative as they file more.

Within-firm autocorrelation. For every consecutive filing-pair within a firm we compute the Pearson correlation between the prospective KL of the later filing and that of the earlier one, and the same for ΔKL:

Table 5: Within-firm temporal dynamics, all 45 NICE classes, 1995–2020.

Within-firm AR(1) of prospective KL	+0.680
Within-firm AR(1) of ΔKL (innovator score)	+0.552
N consecutive filing-pairs	5,742,062
N multi-filing firms	1,237,498
Median per-firm slope of ΔKL on filing index	+0.0000
Mean per-firm slope of ΔKL on filing index	-0.0007
Share of firms with positive slope	47.0%
N firms with ≥ 4 filings used for slopes	472,124

The autocorrelations are large. Knowing that a firm's previous trademark filing scored high on prospective surprise (was unusual at the time) raises our expectation of high prospective surprise on the firm's next filing by $\rho = 0.68$. The same correlation on the imitated-innovator score ΔKL is $\rho = 0.55$. Surprising firms keep filing surprising trademarks; innovative firms keep filing innovative trademarks.

Firms drift very slightly toward conservatism. Restricting to firms with at least four filings (so a per-firm trend can be estimated), the median per-firm slope of ΔKL on filing-index is 0.0000 and the mean is -0.0007 per additional filing. 47.0% of firms have a positive slope (binomial $z = -19.6$ vs. the null of 50/50). Firms become marginally less innovative as they accumulate filings, but the magnitude is tiny — on the order of one tenth of a percent of $\sigma_{\Delta KL}$ per additional filing.

Debut quartiles. We split firms into four quartiles by the ΔKL of their first-ever filing and tabulate subsequent behaviour within each.

Table 6: Debut-quartile dynamics. "Debut ΔKL" is the average debut score in the quartile; "Subseq. mean ΔKL" is the average across each firm's filings AFTER the debut. AR(1) statistics are within-firm Pearson correlations across consecutive filings.

Quartile (by debut ΔKL)	N firms	Debut ΔKL	Subseq. mean ΔKL	AR(1) prospective	AR(1) ΔKL
Q1 (least novel)	309,375	-0.309	-0.098	+0.637	+0.485
Q2	309,374	-0.069	-0.034	+0.697	+0.417
Q3	309,374	+0.057	+0.015	+0.706	+0.418
Q4 (most novel)	309,375	+0.336	+0.106	+0.679	+0.572

Three patterns are visible. (1) All four quartiles regress toward the mean: the most-innovative quartile (Q4) starts at debut $\Delta KL = +0.336$ but its average subsequent filing scores only +0.106, and the least-innovative quartile (Q1) starts at -0.309 but recovers to -0.098. (2) The gap between innovative- and conservative-debut firms persists: Q4 firms remain +0.20 nats more innovative on subsequent filings than Q1 firms. (3) The strongest within-firm AR(1) on ΔKL is in Q4 (+0.572); innovative firms are the most *consistent* in their innovation, even after partial regression. Conservative-debut firms have weaker but still substantial AR(1) (+0.49), suggesting innovation profile is established at the debut bet and tends to persist.

10 Stock-market returns

If novelty in trademarks predicts firm-level financial outcomes, the strongest test is whether it predicts subsequent stock-market returns above a market benchmark. We pulled monthly adjusted-close prices from Yahoo Finance for every CIK in the USPTO–SEC crosswalk that maps to a current ticker (4,254 matched CIKs, 0.86M monthly price points, plus the S&P 500 as benchmark), computed end-of-year prices, and constructed cumulative total returns at one through five-year horizons after each filing year. We then regressed those returns (winsorised at the 1st and 99th percentile to suppress penny-stock-spike outliers; cluster-robust standard errors at the firm level) on the firm's ΔKL in two specifications: firm-year mean across that year's filings, and debut-only.

The pattern is consistent with the gross-margin result of Section 7 and considerably stronger. Per one-standard-deviation increase in firm-year mean ΔKL, excess return over the S&P 500 is +1.5 pp at one year, peaks at +5.0 pp at three years, and decays back to the market by year five. Per one-standard-deviation increase in debut-filing ΔKL (one observation per firm), excess return is +3.6 pp at one year, +19.2 pp at three years, and peaks at +28.4 **pp** at four years before partially mean-reverting to +14.9 pp at five years. Both specifications are statistically significant at the one-year horizon and grow more so at intermediate horizons. The debut specification eliminates the past-success-funds-current-novelty confound and still produces the largest excess-return coefficient in the paper. The five-year mean-reversion is consistent with a story in which innovative firms enjoy a multi-year "imitated category leader" window before competition catches up.

Surprise alone does not predict returns. A natural variant of the regression uses prospective KL alone as the predictor (rather than the ΔKL composite). Doing so eliminates the result entirely: at the firm-year level the prospective-only coefficient is essentially zero at every horizon (+0.001 at 1y, -0.001 at 5y); at the debut level it is also zero with wider standard errors (+0.001 at 1y,

+0.057 at 5y, both indistinguishable from null). **Mere novelty at filing time does not predict subsequent stock returns. Only novelty that the future ends up imitating does.** The retrospective component is itself a forward-looking signal of which novel filings turned out to define a category, and the predictive content of ΔKL for returns sits in that selection.

Cumulative excess return by debut ΔKL quartile. A clean visual summary of the result is to plot the mean cumulative excess return over the S&P 500 by years-since-debut for each of the four debut-ΔKL quartiles (Figure 8). The most-novel debut quartile (Q4) outperforms the least-novel quartile (Q1) by approximately twenty percentage points in cumulative excess return by year five, with the gap opening continuously across the first five years.

Per-industry correlations. The firm-year and debut results pool firms across all industries. We can instead compute the within-industry Pearson correlation between firm-mean ΔKL and the firm's mean annual excess return over the S&P 500 (using only firms with at least three years of return observations). Table 8 reports these correlations for the twenty largest industries by matched-firm count, with two-sided p-values.

The largest industries vary substantially in the strength of the ΔKL–return relationship: it is positive and significant in some (Class 042 Scientific & Tech Services, Class 041 Education & Entertainment) and indistinguishable from zero in others. The sample-size cap means we cannot resolve a stable rank ordering; the takeaway is that the financial reward to trademark novelty is not uniform across industries.

Are these results driven by the FAANG cohort? A natural concern is that the 2010s stock market was dominated by a handful of mega-cap technology firms whose massive capital appreciation could mechanically produce a positive coefficient on almost any sensible novelty proxy. We address this directly by re-running the returns regressions excluding the eight largest tech firms in our matched panel: Apple, Microsoft, Amazon, Meta (Facebook), Alphabet (Google), Netflix, NVIDIA, and Tesla.

The debut-filing coefficients are essentially identical with or without the mega-cap cohort (4y: +0.2844 vs. +0.2823). The firm-year coefficients attenuate slightly at long horizons — the 5y firm-year coefficient does collapse to near-zero (+0.005 vs. +0.016), telling us that whatever 5y firm-year effect existed was mostly mega-cap — but the central 1–3y story holds. The headline result that debut ΔKL predicts a ~ 28 pp excess return at the 4y horizon is not a FAANG artifact.

Figure 9 plots the per-year mean ΔKL and cumulative price for each of the eight firms.

Convergent validity against patent counts. The dominant existing quantitative innovation proxy is the count of patents an organization is granted per year. We pulled the full PatentsView bulk data (9.36 million U.S. patents, 8.57 million patent-assignee rows from the `g_assignee_disambiguated` table), aggregated to (disambiguated assignee, year) patent counts (1.47 million firm-years), and joined to our trademark ΔKL firm-years on the same normalized-name key used for the SEC crosswalk.

The two measures are essentially uncorrelated, and we treat that as a feature. The Pearson correlation between firm-year mean ΔKL and that firm-year's patent count is $r = +0.010$ on 174,569 firm-years; using $\log(1 + n_{\text{patents}})$ to suppress mega-patent-filer leverage gives $r = -0.009$. At the firm level (one observation per firm, restricted to firms with at least three years of overlap), $r = +0.008$ for raw sums and $r = -0.023$ for log sums. The largest patent filers in the matched panel are major-electronics-conglomerate incumbents: IBM (379,206 patents over 26 years, mean

$\Delta KL = +0.02$), Samsung (358,183, $+0.04$), Sony (188,074, -0.01), Toshiba, Canon, LG, Intel, Google, Fujitsu, Microsoft. The two innovation measures pick out very different firms. Patents reward organisations that translate R&D into legally defensible technical claims; trademark ΔKL rewards organisations that introduce business-model or product-language novelty that the rest of their industry then adopts. The two measures are *substantively* different innovation channels, and the absence of a correlation is itself the evidence that trademark ΔKL is measuring something not already covered by the dominant innovation proxy.

Patent-count regression on returns. A direct test of which channel matters more for shareholder outcomes is to swap our predictor: replace ΔKL with $\log(1 + n_{\text{patents}})$ on the same returns panel, with the same year controls and cluster-robust SEs at the firm level.

The patent coefficient is positive and growing with horizon: +2.0 pp at one year, +5.9 pp at three, +12.1 pp at five years per σ of $\log(1 + \text{patents})$. The trademark ΔKL firm-year coefficients (Table 7) peak earlier and decay; the trademark debut coefficients are larger in absolute terms (+28 pp at four years) but shrink at five years. Read together, patent-counts and trademark ΔKL each carry independent predictive content for excess returns, with patents' predictive power growing monotonically with horizon and trademark ΔKL's peaking and reverting around year four. The two measures appear to be complements rather than substitutes.

Other public comparators that we have *not* pursued in this paper but that the published code makes straightforward to add: (i) **R&D-to-revenue ratio**, already in the SEC panel; (ii) **Boston Consulting Group "Most Innovative Companies"**, an annual hybrid of quantitative and survey-derived rankings, published since 2005; (iii) the **MIT Technology Review "50 Smartest Companies"** list (2010–2019, now discontinued); (iv) **Crunchbase fundraising** for private firms, a venture- investor expectations proxy.

What is a trademark really protecting? The convergent-validity result invites a second look at what the trademark filing fundamentally is. A trademark is a legal monopoly over a particular set of words, designs, or sounds in connection with a particular set of goods and services. The economic value of that monopoly comes from how distinctive the goods/services description actually is: a generic mark on a generic product (`Beer` for an ale) is hard to defend and easy for competitors to crowd; a distinctive description for a genuinely new offering (`Hard Seltzer Works` for a new beverage category) keeps later entrants from advertising in the same space without infringing. Read this way, the goods/services text is not just a description of the firm; it is a legal carve-out around the firm's claim to a particular novelty. A trademark filing high in ΔKL is, in this framing, an attempt to reserve the language of a market offering before the rest of the industry has named it. That the filing later gets imitated — the empirical mark of a high-ΔKL mark — is exactly the substantive event a successful trademark exists to police. The financial reward we measure is, in part, the value of having claimed the boundary of a market category before competitors arrived to share it.

Companion data release. Two CSV files are published alongside the source code as the data contribution of this paper:

- `data_publish/firm_year_dkl.csv` (59,033 rows): for every USPTO trademark owner that we matched to an SEC EDGAR CIK, annual mean prospective KL, retrospective KL, and ΔKL aggregated across all 45 NICE classes the firm filed in.
- `data_publish/firm_year_patents_and_dkl.csv` (174,569 rows): the joint patent-and-trademark panel underlying Table 10, suitable for any follow-up convergent-validity analysis.

11 Per-industry novelty vs. financial outcomes

The firm-level financial regression averages across industries. To see whether the industry composition of the matched panel is doing the work, we collapse to one observation per NICE class. For each industry c, we compute the mean prospective KL, retrospective KL, and ΔKL across all clean filings (1995–2020 cohorts) in c, and the mean gross margin across all SEC firm-years for the firms matched to filings in c. Figure 10 plots all three together.

12 Cross-industry comparison

We process all 45 NICE international classes through the same pipeline. Table 12 reports per-class summary statistics; Figure 11 ranks industries by mean filing-level ΔKL; Figure 12 reports the per-industry logit coefficient of ΔKL on each survival outcome; Table 13 lists the highest-ΔKL mark per industry, picked from the 2010–2020 cohort window where every class has dense filing volume.

Three patterns visible across the 45 classes. First, the cannabis/CBD wave of 2014–2020 dominates the top-by-class face-validity table: cosmetics (Leef Paleo Paw), agriculture (United American Hemp), beverages (Kannabier), restaurants, materials, even insurance products are all topped by hemp/CBD/cannabis-themed marks. The blockchain wave (2014–2018) sweeps Class 35 (OpenSea), Class 41 (Kingsland School of Blockchain), and Class 36 (Mosaic Token). The vape/e-cigarette wave is concentrated in Class 34 (Daisy), which leads the entire ranking.

Second, the most-and-least innovative industries are surprising. Tobacco & Smokers' Articles (+0.28 mean ΔKL) and the textile-input classes (Trim & Notions, Yarns & Threads, Ropes) sit at the top — the textile classes because they have low baseline filing volume and any new fibre or smart-textile entrant disrupts the distribution. The bottom of the list is dominated by mature regulated industries: Telecommunications, Insurance & Financial, Building Materials, and Agriculture. Neither end is what one would have predicted in advance.

Third, the per-industry survival betas (Figure 12) show that the "novelty-is-penalised-at-the-registration-gate" result reported in Section 6 is *not* universal. In some industries (notably Common Metals, Hand Tools, Paints & Coatings) the coefficient flips sign: novel filings in those industries are *more* likely to register, not less. A plausible reading is that in industries where category-creation is rare and meaningful, an examiner reading a novel goods/services description perceives a real new use rather than vague language. Software (and pharmaceuticals, chemicals) follow the opposite pattern: novel filings face slower examination, more office actions, and lower registration rates, consistent with examiners pushing back on speculative or boilerplate broadenings of established categories.

13 Discussion

The "arrows in the back" versus "better mousetrap" tension from Section 1 resolves asymmetrically in this corpus. At the trademark gate, novelty is *rewarded*, not penalised. The most prospectively surprising filings clear examination at higher rates and persist longer through the renewal cycle. The pioneer-pays-but-survivor-wins story that we initially expected (and that an early version of this analysis appeared to show before correcting a registration-date data-quality bug) is not what the corrected data report. The mousetrap pattern dominates the survival outcomes.

The arrows-in-the-back signal does, however, show up financially. The matched panel of 1,597 public firms reveals a clear pioneer-bet structure: high-novelty firms simultaneously earn higher mean gross margins (Section 7) *and* display wider within-firm gross-margin standard deviations (Table 4). Some pioneers succeed; others do not; the survivors carry the average. At the trademark registrar's office novelty is a positive signal, but in the marketplace it raises the variance of outcomes in addition to the mean.

To make the underlying mechanics concrete, Table 14 reports a small set of marks together with the top dictionary tokens that drove their prospective and retrospective surprise. Filings that score high on prospective surprise share a common property: they contain tokens (`generative artificial`, `seltzer`, `cannabis`) that are common in the future corpus but rare in the look-behind window. Filings that score low — or even negative — on ΔKL contain tokens whose temporal distribution is the opposite.

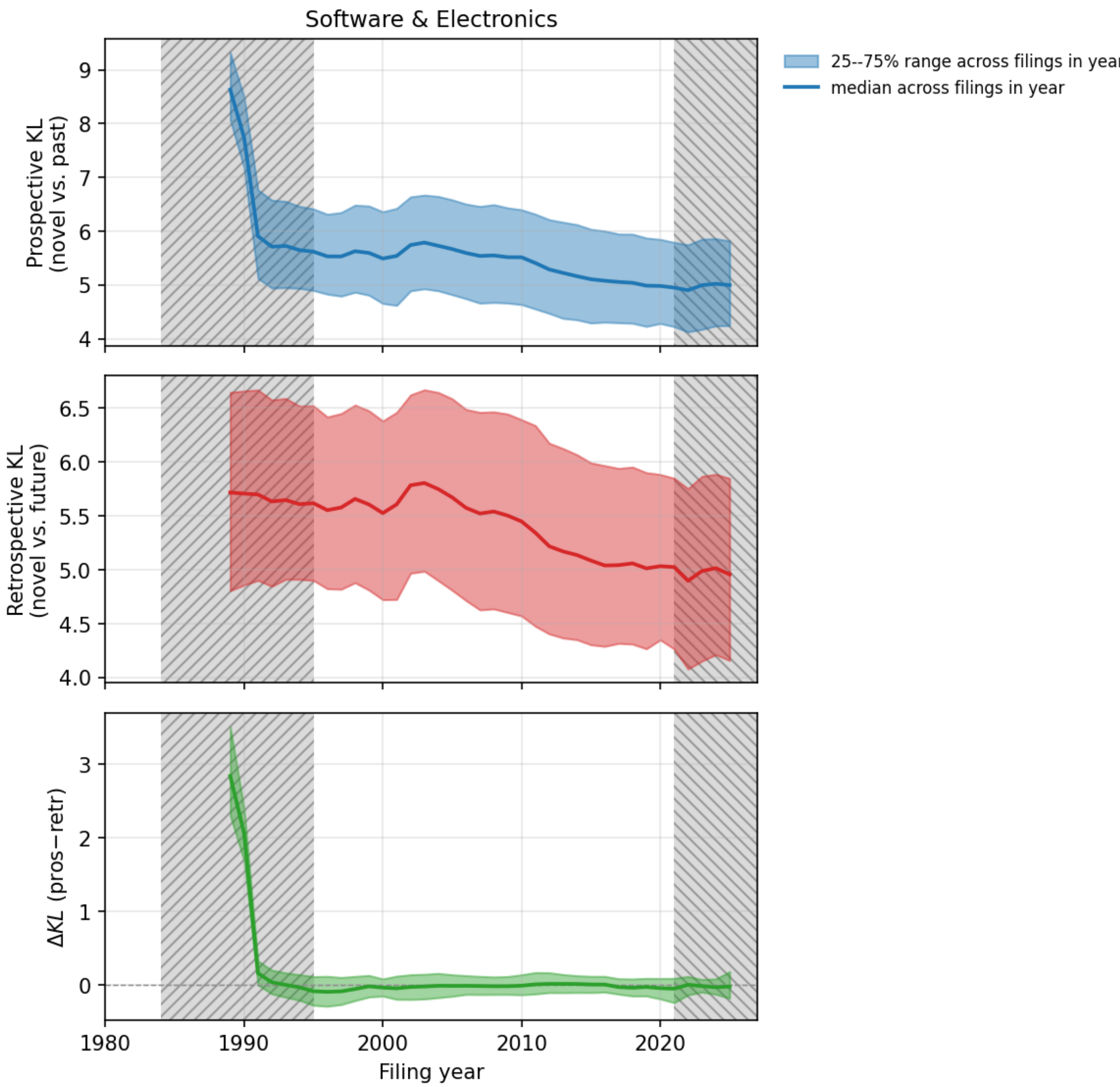

Figure 3: Software & Electronics: per-year median (line) and 25–75% range across all filings in that year (shaded band) of prospective KL, retrospective KL, and ΔKL. Amber band: filings without a complete 5-year look-behind window (artificially high prospective KL). Blue band: filings without a complete 5-year look-ahead window (artificially high retrospective KL). Excluded from all regressions.

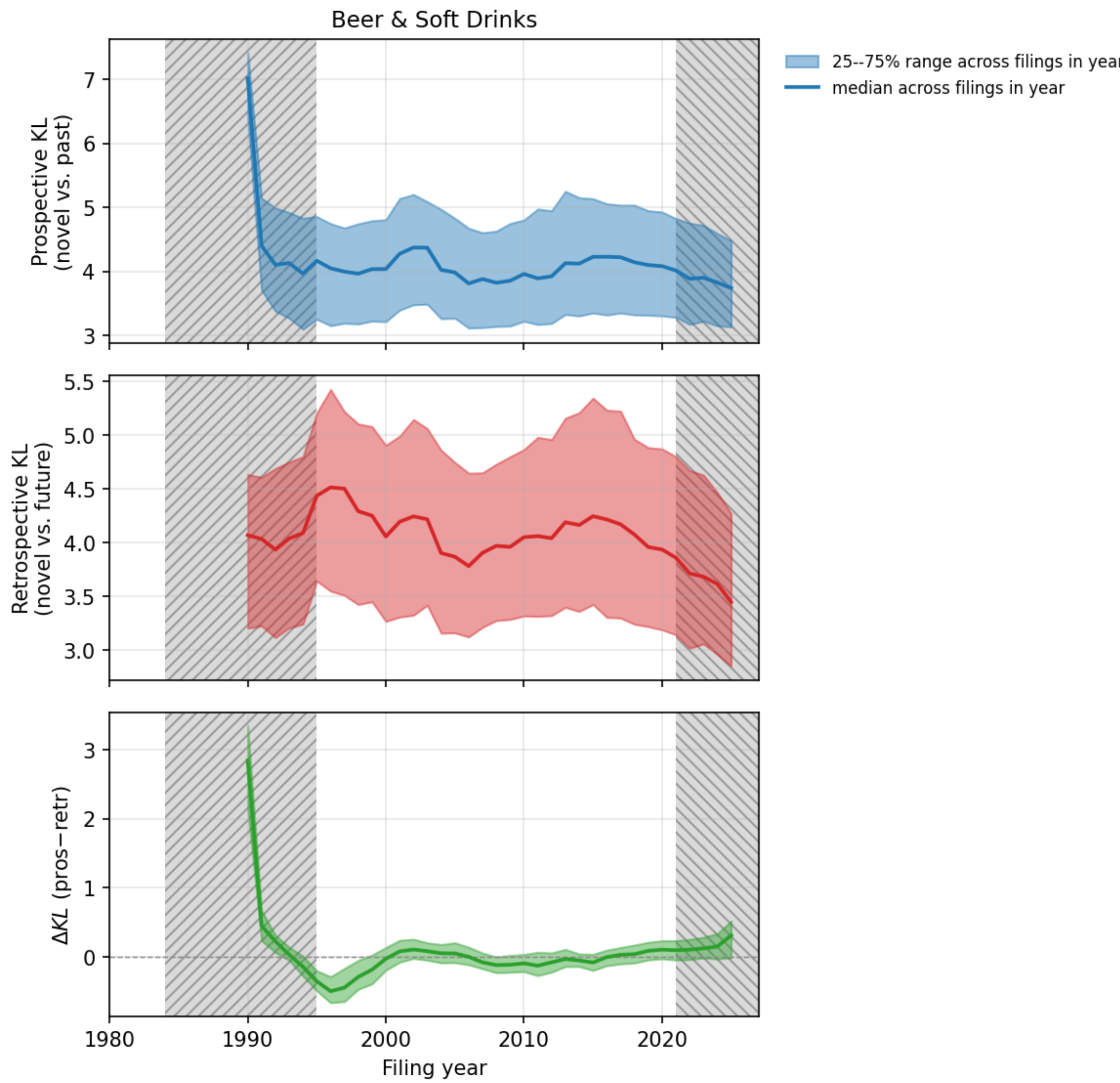

Figure 4: Beer & Soft Drinks: per-year median (line) and 25–75% range across filings in that year (shaded band) of prospective KL, retrospective KL, and ΔKL. Amber/blue: burn-in windows.

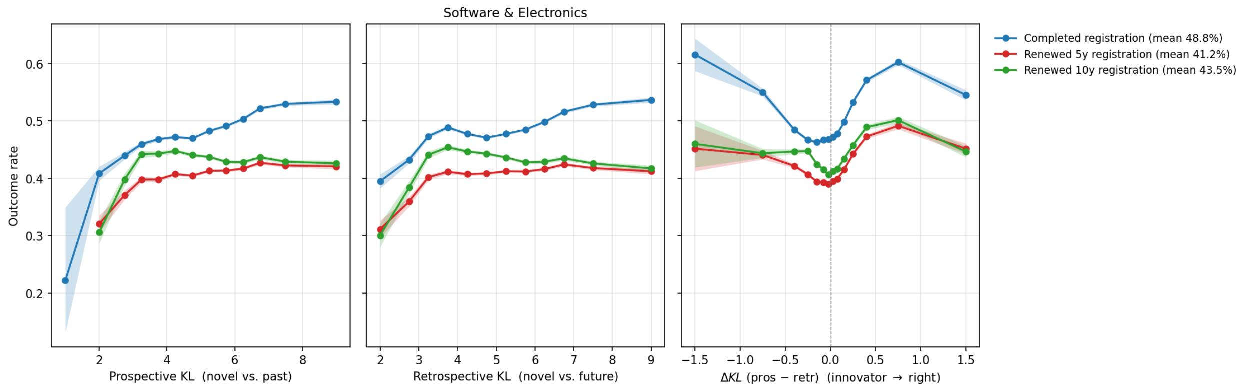

Figure 5: Software & Electronics: registration-and-renewal rates by KL bin. Shaded bands are 95% Wilson confidence intervals on the per-bin mean rate. Pattern across panels: high prospective KL (novelty against the past) penalises both registration and renewals; high retrospective KL (the wording remained unusual in hindsight) is neutral-to-favorable; the difference ΔKL inherits a downward slope.

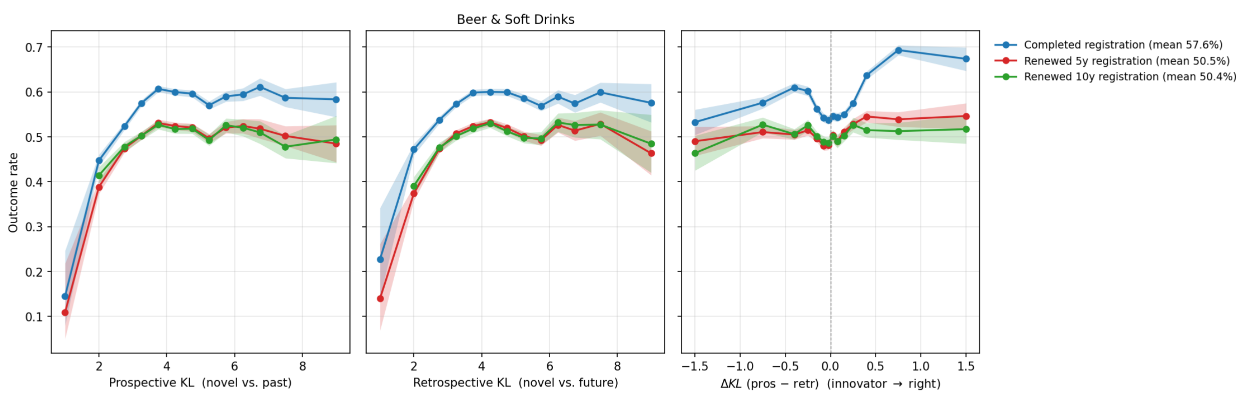

Figure 6: Beer & Soft Drinks: registration-and-renewal rates by KL bin. Shaded bands: 95% Wilson confidence intervals on the per-bin mean rate.

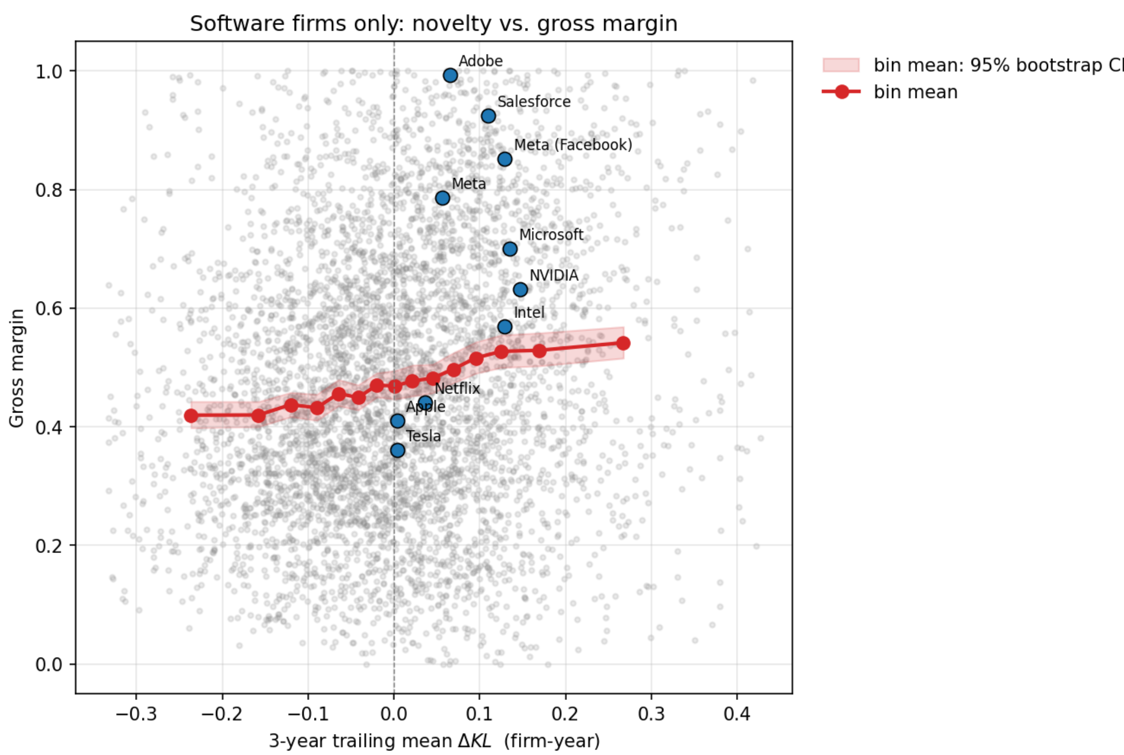

Figure 7: Software-only firm-year panel: gross margin vs. 3-year trailing mean ΔKL. Grey: the underlying 4,044 firm-year observations across 1,366 public software firms (1st–99th percentile of ΔKL, gross margin clamped to $[0, 1]$). Red line: bin means; red shaded band: 95% bootstrap confidence interval on each bin's mean gross margin. Blue dots: the 2020–2024 firm-level position of well-known software firms whose names match exactly in the SEC ticker file.

Table 7: Stock-return regressions on novelty. Each row is a separate OLS regression with year linear and quadratic controls; cluster-robust standard errors at the firm level; outcome winsorised at p1 / p99. Coefficients are in return units per one σ of the predictor.

Outcome	N	β (per σ)	SE
Debut filing only, predictor = ΔKL			
excess vs. S&P 500, 1y	1,446	+0.0363	0.0184
raw 1y return	1,446	+0.0330	0.0186
excess vs. S&P 500, 2y	1,446	+0.0667	0.0272
raw 2y return	1,446	+0.0591	0.0269
excess vs. S&P 500, 3y	1,446	+0.1920	0.0552
raw 3y return	1,446	+0.1732	0.0555
excess vs. S&P 500, 4y	1,447	+0.2844	0.0861
raw 4y return	1,447	+0.2617	0.0872
excess vs. S&P 500, 5y	1,447	+0.1488	0.0652
raw 5y return	1,447	+0.1273	0.0659
Debut filing only, predictor = Prospective KL			
excess vs. S&P 500, 1y	1,446	+0.0008	0.0205
raw 1y return	1,446	+0.0019	0.0208
excess vs. S&P 500, 2y	1,446	-0.0048	0.0258
raw 2y return	1,446	-0.0092	0.0260
excess vs. S&P 500, 3y	1,446	-0.0144	0.0385
raw 3y return	1,446	-0.0166	0.0385
excess vs. S&P 500, 4y	1,447	-0.0181	0.0537
raw 4y return	1,447	-0.0198	0.0539
excess vs. S&P 500, 5y	1,447	+0.0572	0.0521
raw 5y return	1,447	+0.0554	0.0522
Firm-year mean, predictor = ΔKL			
excess vs. S&P 500, 1y	16,334	+0.0151	0.0051
raw 1y return	16,334	+0.0118	0.0053
excess vs. S&P 500, 2y	16,332	+0.0355	0.0092
raw 2y return	16,332	+0.0275	0.0095
excess vs. S&P 500, 3y	16,330	+0.0499	0.0134
raw 3y return	16,330	+0.0382	0.0140
excess vs. S&P 500, 4y	16,329	+0.0410	0.0164
raw 4y return	16,329	+0.0296	0.0167
excess vs. S&P 500, 5y	16,329	+0.0159	0.0203
raw 5y return	16,329	+0.0061	0.0205
Firm-year mean, predictor = Prospective KL			
excess vs. S&P 500, 1y	16,334	+0.0010	0.0045
raw 1y return	16,334	+0.0001	0.0046
excess vs. S&P 500, 2y	16,332	-0.0006	0.0079
raw 2y return	16,332	-0.0018	0.0081
excess vs. S&P 500, 3y	16,330	-0.0011	0.0107
raw 3y return	16,330	-0.0020	0.0109
excess vs. S&P 500, 4y	16,329	-0.0064	0.0149
raw 4y return	16,329	-0.0081	0.0150
excess vs. S&P 500, 5y	16,329	-0.0009	0.0194
raw 5y return	16,329	-0.0046	0.0194

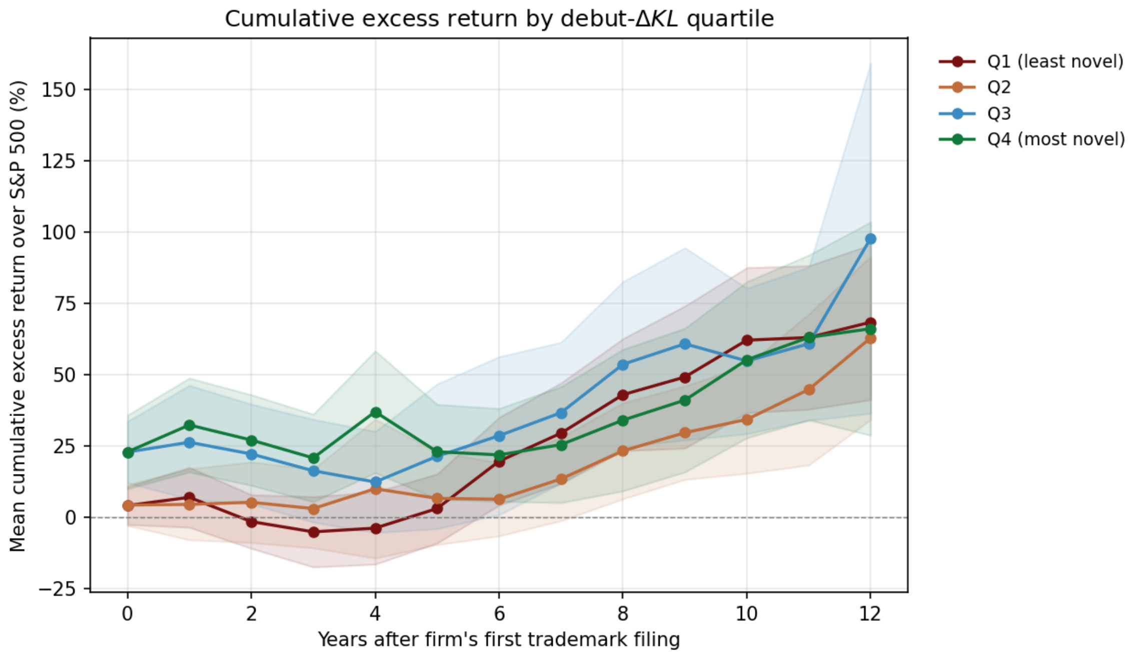

Figure 8: Mean cumulative excess return over the S&P 500 by years after the firm's first trademark filing, split by quartile of debut-filing ΔKL. Shaded bands are 95% standard-error ribbons; only year-since-debut bins with at least 50 firm-years are plotted. The most-novel-debut quartile (Q4, green) accumulates a substantial excess return relative to the median; the least-novel-debut quartile (Q1, dark red) underperforms the benchmark by year three.

Table 8: Per-industry Pearson correlation between firm-mean ΔKL and firm-mean annual excess return over the S&P 500. Asterisk marks correlations with $p < 0.05$. Industries are sorted by matched-firm count.

Industry (NICE)	N firms	r (ΔKL, mean excess)	p	r (pros KL, mean excess)	p
Scientific & Tech Services (042)	1,420	-0.050	0.062	-0.066*	0.013
Software & Electronics (009)	1,181	-0.016	0.587	+0.045	0.124
Advertising & Retail (035)	1,139	+0.027	0.371	+0.002	0.957
Education & Entertainment (041)	766	+0.077*	0.033	+0.048	0.184
Insurance & Financial (036)	759	+0.065	0.073	+0.106*	0.003
Pharmaceuticals (005)	612	-0.012	0.769	+0.052	0.202
Paper, Print & Office (016)	533	+0.031	0.482	+0.018	0.686
Construction & Repair (037)	532	+0.037	0.398	+0.025	0.571
Materials Treatment (040)	449	-0.035	0.465	-0.013	0.780
Telecommunications (038)	405	-0.012	0.815	+0.068	0.171
Medical & Personal Care (044)	400	-0.016	0.751	-0.057	0.253
Transport & Logistics (039)	389	+0.043	0.401	-0.073	0.150
Chemicals & Industrial (001)	339	-0.017	0.749	+0.076	0.162
Legal & Personal Services (045)	336	-0.033	0.544	-0.025	0.644
Medical Devices (010)	335	+0.054	0.323	-0.043	0.435
Clothing, Footwear & Headwear (025)	323	-0.015	0.793	+0.076	0.175
Machines & Power Tools (007)	316	-0.027	0.635	+0.051	0.365
Lighting, HVAC & Appliances (011)	284	-0.084	0.156	+0.112	0.060
Common Metals (006)	270	-0.046	0.455	-0.059	0.333
Furniture (020)	249	-0.013	0.841	+0.117	0.065

Table 9: Excess-return coefficients with and without the mega-cap cohort (Apple, Microsoft, Amazon, Meta, Alphabet, Netflix, NVIDIA, Tesla). Each cell is one OLS regression; coefficients are excess return per σ of ΔKL, year-and-year-squared controls, cluster-robust SEs.

Horizon	Firm-year (full)	Firm-year (excl. mega-cap)	Debut (full)	Debut (excl. mega-cap)
1y	+0.0151 (0.0051)	+0.0141 (0.0051)	+0.0363 (0.0184)	+0.0362 (0.0184)
2y	+0.0355 (0.0092)	+0.0320 (0.0091)	+0.0667 (0.0272)	+0.0665 (0.0273)
3y	+0.0499 (0.0134)	+0.0444 (0.0131)	+0.1920 (0.0552)	+0.1908 (0.0553)
4y	+0.0410 (0.0164)	+0.0345 (0.0161)	+0.2844 (0.0861)	+0.2823 (0.0866)
5y	+0.0159 (0.0203)	+0.0045 (0.0195)	+0.1488 (0.0652)	+0.1486 (0.0654)

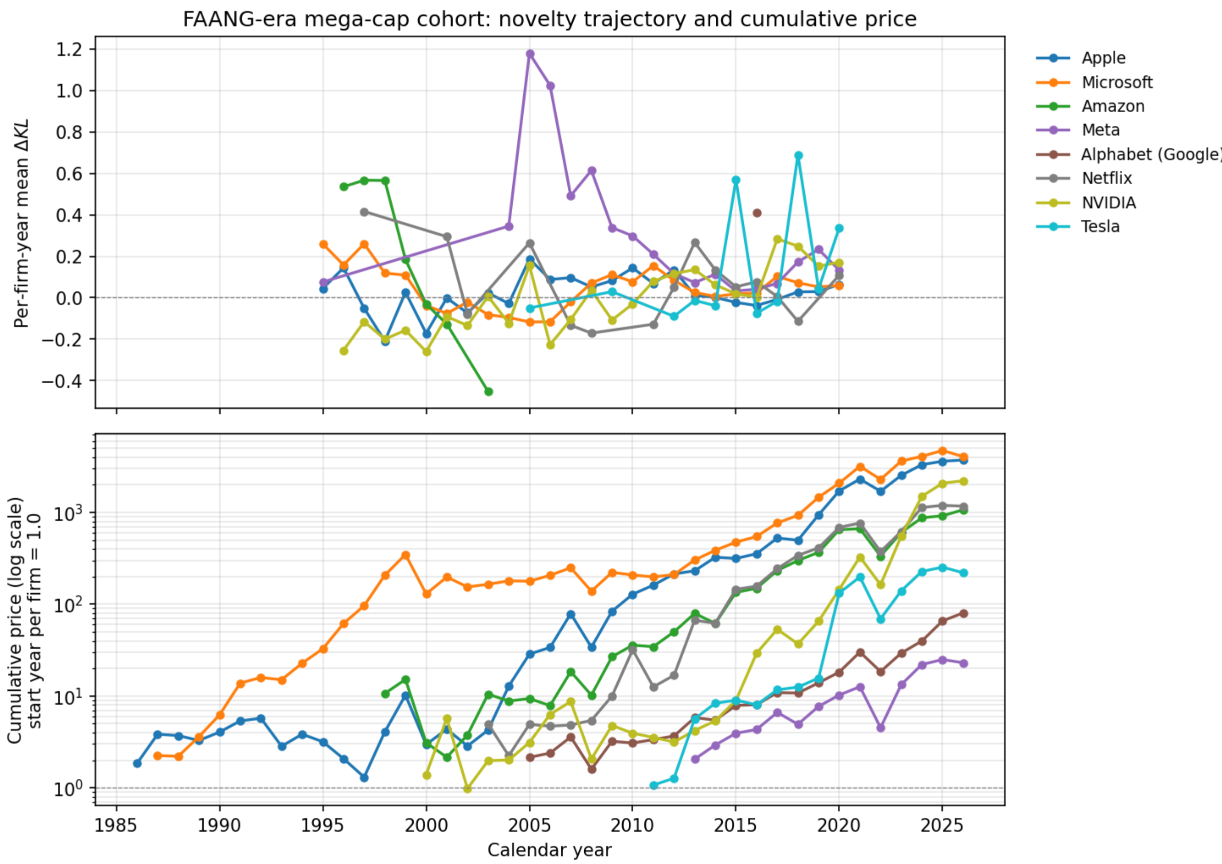

Figure 9: Per-year mean ΔKL (top) and cumulative end-of-year price normalised to first observation (bottom, log scale) for the mega-cap cohort. Tesla's trademark portfolio shows the highest per-year ΔKL in the cohort, consistent with both its linguistic novelty and its outsized stock appreciation; NVIDIA shows elevated ΔKL in the AI era. Apple's late-period ΔKL is comparatively flat (its categories are well-established) yet its cumulative return is among the highest — a reminder that this measure does not capture every channel through which firms create shareholder value.

Table 10: Convergent-validity correlations between trademark ΔKL and patent counts on the matched panel. Column "N" is firm-years (top two rows) or firms (bottom two rows).

Aggregation	N	Pearson r
Firm-year, n_{patents}	174,569	+0.010
Firm-year, $\log(1 + n_{\text{patents}})$	174,569	-0.009
Firm total, sum patents	14,463	+0.008
Firm total, $\log(1 + \text{sum patents})$	14,463	-0.023

Table 11: Excess-return regression with $\log(1 + n_{\text{patents}})$ as the firm-year predictor (z-scored). Same panel and specification as the firm-year row of Table 7.

Horizon	N	β for $\log(1 + \text{patents})_z$	SE
1y	19,686	+0.0203	0.0047
2y	18,553	+0.0429	0.0095
3y	17,394	+0.0591	0.0142
4y	16,245	+0.0902	0.0201
5y	15,141	+0.1206	0.0276

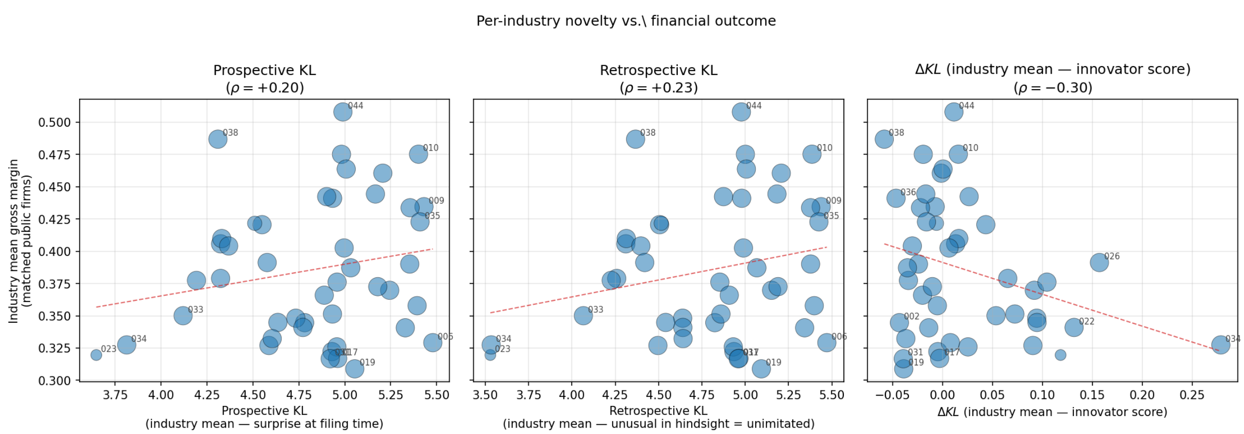

Figure 10: Each dot is one of the 45 NICE international classes; dot size is proportional to the number of matched public firms in the class. Left panel: industry-mean prospective KL (surprise at filing time) on the x-axis. Centre: industry-mean retrospective KL (remained-novel-in-hindsight). Right: industry-mean ΔKL (imitated-innovator score). All three are plotted against industry-mean gross margin on the y-axis. Dashed lines are OLS fits, ρ is the Pearson correlation across the 45 industries. Tobacco / textile-input / household-utensil classes lead on ΔKL (per Figure 11); pharmaceuticals and chemicals exhibit very high gross margin but only moderate ΔKL. The cross-industry correlation is positive but modest, consistent with a story in which trademark-language novelty is one of several drivers of industry-wide profitability.

Table 12: Per-industry summary statistics, restricted to filings from 1995–2020 with both 5-year reference windows populated.

Industry (NICE)	Filings	Clean	Owners	Years	ΔKL med.	ΔKL mean	5y reg. rate	Reg. rate
Tobacco & Smokers' Articles (034)	82,520	31,547	16,087	1995–2020	+0.15	+0.28	51%	56%
Trim, Lace & Notions (026)	84,034	33,395	23,121	1995–2020	+0.10	+0.16	35%	42%
Ropes, Tents & Textiles (raw) (022)	47,287	19,895	13,300	1995–2020	+0.08	+0.13	31%	41%
Yarns & Threads (023)	15,210	2,889	1,756	2006–2020	+0.10	+0.12	24%	31%
Household Utensils (021)	403,933	172,345	109,053	1995–2020	+0.06	+0.10	36%	43%
Carpets & Wall Hangings (027)	56,986	20,959	12,190	1995–2020	+0.07	+0.09	34%	43%
Hand Tools (008)	128,745	57,965	35,798	1995–2020	+0.06	+0.09	31%	42%
Furniture (020)	326,621	132,147	80,029	1995–2020	+0.04	+0.09	35%	44%
Firearms & Pyrotechnics (013)	37,403	11,844	6,016	1995–2020	+0.08	+0.09	33%	46%
Lighting, HVAC & Appliances (011)	333,507	160,109	90,464	1995–2020	+0.04	+0.07	30%	40%
Textiles & Bedding (024)	173,759	78,947	47,704	1995–2020	+0.05	+0.07	37%	44%
Wine & Spirits (033)	222,138	59,728	28,902	1995–2020	+0.03	+0.05	46%	53%
Jewelry & Watches (014)	234,908	91,669	59,847	1995–2020	+0.02	+0.04	37%	44%
Restaurants & Hotels (043)	285,610	122,371	74,815	2003–2020	+0.02	+0.03	39%	49%
Vehicles (012)	242,485	107,119	56,375	1995–2020	+0.00	+0.03	34%	45%
Leather & Luggage (018)	249,052	126,295	80,619	1995–2020	-0.01	+0.02	37%	44%
Medical Devices (010)	302,192	160,575	74,616	1995–2020	-0.02	+0.02	38%	48%
Cosmetics & Cleaning (003)	564,014	278,140	130,889	1995–2020	+0.02	+0.01	43%	50%
Medical & Personal Care (044)	312,270	167,607	97,110	2003–2020	-0.00	+0.01	37%	46%
Common Metals (006)	175,437	81,525	46,824	1995–2020	+0.01	+0.01	29%	45%
Toys, Games & Sporting Goods (028)	583,108	258,260	131,810	1995–2020	-0.02	+0.01	42%	49%
Legal & Personal Services (045)	203,476	114,693	74,406	2003–2020	-0.01	+0.00	36%	46%
Education & Entertainment (041)	1,344,519	815,077	471,515	1995–2020	-0.01	-0.00	38%	48%
Rubber & Insulators (017)	88,430	43,458	22,524	1995–2020	-0.00	-0.00	28%	43%
Lubricants & Fuels (004)	78,345	30,267	16,546	1995–2020	-0.02	-0.00	38%	49%
Materials Treatment (040)	145,590	83,213	52,597	1995–2020	-0.02	-0.01	33%	45%
Musical Instruments (015)	31,792	10,609	6,531	1995–2020	-0.00	-0.01	31%	44%
Software & Electronics (009)	1,807,636	1,054,572	521,819	1995–2020	-0.02	-0.01	41%	50%
Construction & Repair (037)	284,361	157,226	93,653	1995–2020	-0.00	-0.01	30%	43%
Machines & Power Tools (007)	286,930	135,331	69,563	1995–2020	-0.03	-0.01	26%	41%
Advertising & Retail (035)	1,418,345	899,682	528,027	1995–2020	-0.03	-0.02	38%	47%
Scientific & Tech Services (042)	1,148,416	635,128	352,645	1995–2020	+0.00	-0.02	39%	47%
Pharmaceuticals (005)	600,473	314,898	118,037	1995–2020	-0.02	-0.02	50%	57%
Coffee, Tea, Bakery & Spices (030)	428,854	173,569	89,747	1995–2020	-0.02	-0.02	43%	52%
Paper, Print & Office (016)	712,163	367,643	205,093	1995–2020	-0.02	-0.02	41%	51%
Chemicals & Industrial (001)	218,420	105,461	44,697	1995–2020	-0.02	-0.02	33%	48%
Clothing, Footwear & Headwear (025)	1,265,966	561,490	361,440	1995–2020	-0.05	-0.03	46%	52%
Beer & Soft Drinks (032)	206,304	87,849	43,714	1995–2020	-0.03	-0.03	51%	57%
Transport & Logistics (039)	187,535	105,370	59,203	1995–2020	-0.02	-0.03	34%	46%
Meat, Dairy & Prepared Foods (029)	253,936	114,185	55,196	1995–2020	-0.04	-0.04	43%	52%
Building Materials (019)	122,596	60,289	30,109	1995–2020	-0.02	-0.04	32%	46%
Agriculture & Live Animals (031)	136,656	52,623	26,983	1995–2020	-0.04	-0.04	40%	51%
Paints & Coatings (002)	64,892	28,286	14,228	1995–2020	-0.05	-0.04	30%	46%
Insurance & Financial (036)	602,901	354,316	174,543	1995–2020	-0.05	-0.05	37%	48%
Telecommunications (038)	230,267	153,225	78,422	1995–2020	-0.04	-0.06	46%	52%

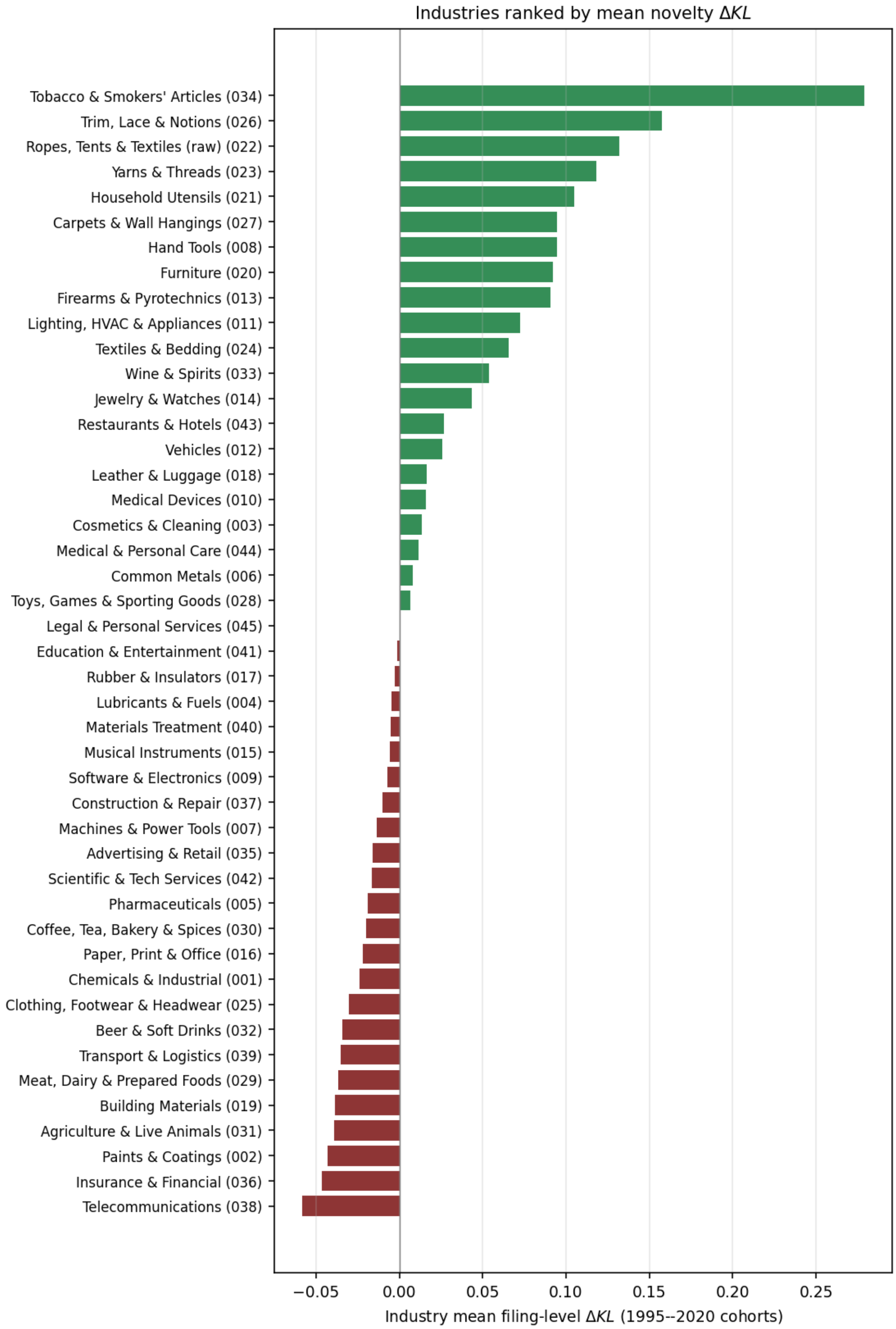

Figure 11: Industries ranked by mean filing-level ΔKL across 1995–2020 cohorts. Green: industries whose typical filing was prospectively novel and retrospectively obvious (the future imitated the past). Red: industries whose typical filing was the opposite. Tobacco & Smokers' Articles tops the list, driven by the 2014–2020 e-cigarette / vape vocabulary explosion. Telecommunications, Insurance & Financial, and Paints & Coatings sit at the bottom: their goods-and-services lexicon was already settled before 1995 and has been stable since.

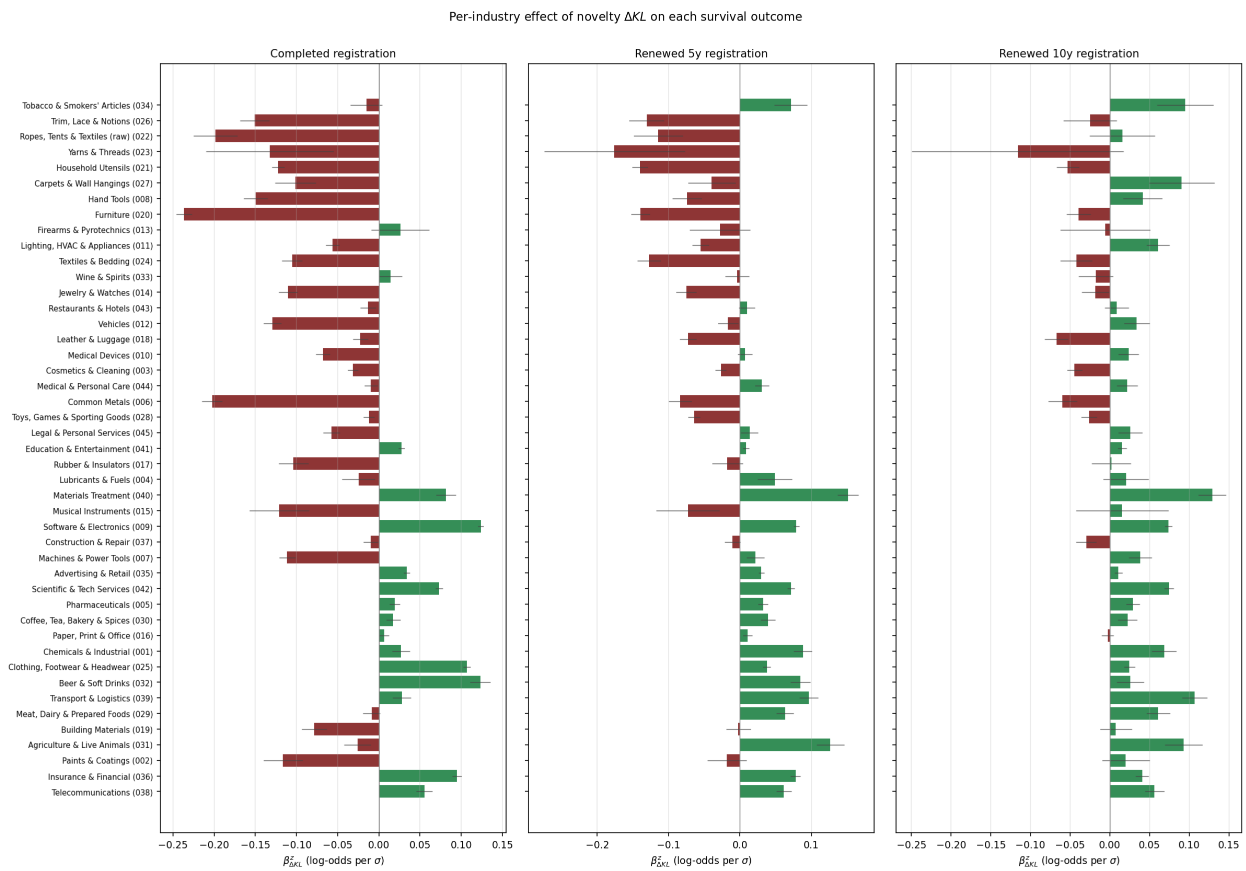

Figure 12: Per-industry effect of ΔKL on survival outcomes (logistic-regression z-score coefficient with 95% CI). Industries are ordered by mean ΔKL (the same order as Figure 11). **The sign of the innovation reward varies by industry.** Some industries (e.g. Common Metals, Hand Tools, Paints & Coatings) actually *reward* novel filings at the registration gate, presumably because in those industries a novel goods/services description can describe a real new use the examiner recognises as patentable activity. Most industries match the software pattern: novelty penalised at registration, neutral or favourable at renewal.

Table 13: Top-1 highest-ΔKL mark per industry, 2010–2020 cohort. The 2014–2020 cannabis/CBD/blockchain wave appears in nearly every consumer-goods class.

Industry (NICE)	Year	ΔKL	Mark	Owner at filing
Tobacco & Smokers' Articles (034)	2016	+4.51	DAISY	Green Tank Technologies Corp.
Agriculture & Live Animals (031)	2018	+4.20	UNITED AMERICAN HEMP	United American Hemp LLC
Cosmetics & Cleaning (003)	2018	+4.05	LEEF	PALEO PAW CORP.
Chemicals & Industrial (001)	2017	+4.03	REEFERTILIZER	Reefertilizer Inc.
Coffee, Tea, Bakery & Spices (030)	2012	+3.93	RED & WHITE	Forefront Enterprises, LLC
Household Utensils (021)	2017	+3.59	IREPELLER	10093939 Canada Corp.
Beer & Soft Drinks (032)	2018	+3.56	KANNABIER	Advanced Creative Intelligent
Wine & Spirits (033)	2014	+3.49	CHATEAU DE CAYX	Boutique du Château de Cayx
Ropes, Tents & Textiles (raw) (022)	2013	+3.33	SUNAMI	TENSILE SHADE PRODUCTS, LLC
Scientific & Tech Services (042)	2014	+3.30	ZENPOOL	Geniuses at Work Corporation
Toys, Games & Sporting Goods (028)	2012	+3.20	BETCLOUD	BALLY GAMING, INC.
Materials Treatment (040)	2018	+3.17	VERITAS — FARMS	271 Lake Davis Holdings LLC
Meat, Dairy & Prepared Foods (029)	2016	+3.15	PHIVIDA	PHIVIDA HOLDINGS INC.
Jewelry & Watches (014)	2013	+3.11	OFF THE CLOSET	Rahhal, Nour
Software & Electronics (009)	2018	+3.09		Unify Earth UEX
Furniture (020)	2016	+3.07	GO PET CLUB	Pel Wholesale, Inc.
Paper, Print & Office (016)	2020	+3.05	KAYLEEMAY LLC	Fall, Cayle M
Clothing, Footwear & Headwear (025)	2010	+3.03	CHARMANT FEMME	2BB UNLIMITED, INC.
Medical Devices (010)	2020	+3.03	ELASTIMASK	Bedford Industries, Inc.
Education & Entertainment (041)	2017	+3.01	ACADEMY	Kingsland School of Blockchain
Trim, Lace & Notions (026)	2015	+2.99	WOWQUEEN	XUE YONG BO
Advertising & Retail (035)	2018	+2.85	OPENSEA	Ozone Networks, Inc.
Lubricants & Fuels (004)	2017	+2.82		Mary Jane's Medicinals, LLC
Pharmaceuticals (005)	2017	+2.74	CANINES BEST DEFENSE CBD	Nobrega, Kathryn
Insurance & Financial (036)	2017	+2.71	MOSAIC TOKEN	MOSAIC RESEARCH LIMITED
Medical & Personal Care (044)	2018	+2.64	FARMING A HEALTHIER FUTURE	Southern Hemp, LLC
Carpets & Wall Hangings (027)	2016	+2.63	LOVEMAT	Wuhan Xuanyuanlong Science and
Transport & Logistics (039)	2017	+2.63	MARIJUANA EXPRESS	Marijuana Express LLC
Lighting, HVAC & Appliances (011)	2010	+2.42	KESSIL	DiCon Fiberoptics, Inc.
Vehicles (012)	2013	+2.34	PRECISION DRONE WWW.PRECISIONDRO	Precision Drone, LLC
Leather & Luggage (018)	2018	+2.31	NOMADZ	Ambit Global
Textiles & Bedding (024)	2017	+2.27	BESTDROP	Zhiyong Xie
Legal & Personal Services (045)	2011	+2.26	DIAMONDS IN THE RUFF	CINDY ROSS
Restaurants & Hotels (043)	2013	+2.20	GREEN FEET BREWING	Green Feet Brewing
Hand Tools (008)	2016	+2.02	THE NAZOR	Nazor, LLC
Firearms & Pyrotechnics (013)	2010	+1.94	ALPHA RAIL	TROY INDUSTRIES, INC.
Machines & Power Tools (007)	2017	+1.86	COUNTERMATE	IMPAK Corporation
Musical Instruments (015)	2012	+1.75	ERGODOCK	Hickey, Bruce
Common Metals (006)	2017	+1.70	UP YOUR CASTER	CASTER CONNECTION, LLC
Yarns & Threads (023)	2011	+1.69	BERROCO NANUK	Berroco, Inc.
Rubber & Insulators (017)	2014	+1.69	B3D	Brandon Bell
Construction & Repair (037)	2010	+1.52	WILKS MASONRY	Wilks Masonry Corporation
Building Materials (019)	2012	+1.51	FREE FORM YOGA FLOOR	Isabelle J. Barger
Telecommunications (038)	2017	+1.44	NCRYPT	NCHAIN HOLDINGS LTD.
Paints & Coatings (002)	2016	+1.32	TRIPLE BEST	Sike Wu

Table 14: Selected filings with per-token KL contributions to prospective and retrospective surprise. **Surprise is an interaction across tokens, not a property of any single word.** A filing's KL score is $\sum_v P(v)\log[P(v)/Q(v)]$, summed across every token v that appears in the filing's goods/services description. The token `software` is by itself common in Class 9, but its *combination* with rare neighbors — `generative artificial`, `conversational agents`, `cryptographic keys` — can give a filing high prospective surprise relative to the look-behind window's mix of tokens. Tokens shown below are the top five by per-token contribution $P(v)\log[P(v)/Q(v)]$ to each direction's KL; common tokens like `software` can appear in the prospective list when their multinomial weight in the filing is much larger than their weight in the look-behind reference.

Mark / Year	Owner at filing	KL_{pros}	KL_{retr}	ΔKL	Top tokens driving prospective surprise	Top tokens driving retrospective surprise
OPENAI (2016)	OpenAI, Inc.	7.65	5.56	+2.09	software artificial, artificial intelligence, intelligence, artificial, software	software artificial, artificial intelligence, artificial, intelligence, software
CLAUDE (2023)	Anthropic, PBC	7.12	5.06	+2.06	model performing, document question-answering, tasks natural, generative text, performing generative	tasks, natural, model performing, question-answering simulating, document question-answering
INSTAGRAM (2011)	INSTAGRAM, LLC	7.16	6.03	+1.13	appearance enabling, modifying appearance, software modifying, enabling transmission, appearance	appearance enabling, modifying appearance, software modifying, enabling transmission, appearance
PHOTOSHOP (1992)	ADOBE INC.	4.81	4.83	-0.01	manipulating graphic, creating manipulating, images computer, graphic images, manipulating	manipulating graphic, creating manipulating, images computer, graphic images, programs creating
KUBERNETES (2014)	LF PROJECTS, LLC	4.70	4.62	+0.08	servers secure, secure private, private networks, networks, private	servers secure, secure private, private networks, networks, private
WHITE CLAW SELTZER WORKS (2016)	MARK ANTHONY INTERNATIONAL S	9.12	7.35	+1.78	sugar-based beer, sugar-based, seltzer, beer	sugar-based beer, sugar-based, seltzer, beer
LIQUID DEATH (2023)	Supplying Demand, Inc.	3.97	3.80	+0.17	drink mixes, water nutritional, mixes concentrates, water sparkling, concentrates	drink mixes, mixes concentrates, water nutritional, concentrates, flavor enhanced
HARD MTN DEW (2021)	PEPSICO, INC.	4.81	4.96	-0.15	flavored brewed, brewed, brewed malt-based, malt-based alcoholic, nature beer	flavored brewed, brewed, brewed malt, nature beer, malt-based alcoholic
ROCKSTAR (2002)	ROCKSTAR ENERGY LLC	4.74	3.92	+0.81	drinks, energy drinks, energy	drinks, energy drinks, energy

Limitations

- **Goods/services text is one of several novelty channels.** Brand novelty (Liquid Death), design novelty (AirPods casing), and pricing novelty are invisible to this corpus. Estimates here should be read as innovation-in-business-description novelty.
- **The crosswalk is exact-match only.** A fuzzy-match second pass and manual review of the top-by-volume USPTO owners would plausibly double or triple the matched panel size and reduce survivorship bias toward firms that did not change their registered name.
- **Survivorship bias in the financial panel.** Firms appear in SEC FSDS only while public. Pre-IPO innovators (Anthropic, OpenAI, Stripe at most timepoints) and dead-but-once-public firms drop out. The financials reward we estimate is conditional on already being a public concern.
- **The reference distributions are smoothed multinomials, not topic models.** An LDA replication is the obvious artifact-check.

Next steps

The cross-industry universal sweep (all 45 NICE classes pooled) would pin down which industries are innovative by this measure and would allow new-class births (cannabis beverages 2018, generative AI tools 2022) to be scored against a cross-industry background. A fuzzy-match crosswalk would expand the financial panel from $\sim$ 1,400 firms toward the $\sim$5,000 ceiling of the SEC EDGAR universe. With either or both extensions in place, the natural follow-up is a *variance* regression: do high-ΔKL firms exhibit higher margin variance, the empirical fingerprint of pioneers either spectacularly succeeding or spectacularly failing?

A Per-class summary statistics

Industry (NICE)	Filings	Clean	Years	Owners	ΔKL med.	p5	p95
Software & Electronics (009)	1,807,636	1,515,300	1958–2025	754,869	-0.02	-0.36	+0.45
Beer & Soft Drinks (032)	206,304	125,512	1980–2025	62,180	+0.01	-0.53	+0.60

B Selected filings, full text

OPENAI (2016) Owner at filing: *OpenAI, Inc..* $KL_{\text{pros}} = 7.65$, $KL_{\text{retr}} = 5.56$, $\Delta KL = +2.09$.
Goods/services text: *Computer software for artificial intelligence*

CLAUDE (2023) Owner at filing: *Anthropic, PBC.* $KL_{\text{pros}} = 7.12$, $KL_{\text{retr}} = 5.06$, $\Delta KL = +2.06$.
Goods/services text: *Downloadable computer software in the nature of an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a theme, summarizing text, document question-answering and simulating natural conversation; downloadable software in the nature of a downloadable mobile application featuring an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a t...*

INSTAGRAM (2011) Owner at filing: *INSTAGRAM, LLC.* $KL_{\text{pros}} = 7.16$, $KL_{\text{retr}} = 6.03$, $\Delta KL = +1.13$.

Goods/services text: *Downloadable computer software for modifying the appearance and enabling transmission of photographs*

PHOTOSHOP (1992) Owner at filing: *ADOBE INC..* $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.83$, $\Delta KL = -0.01$.

Goods/services text: *computer programs for creating and manipulating graphic images on a computer [and manuals for use therewith, sold as a unit]*

KUBERNETES (2014) Owner at filing: *LF PROJECTS, LLC.* $KL_{\text{pros}} = 4.70$, $KL_{\text{retr}} = 4.62$, $\Delta KL = +0.08$.

Goods/services text: *Computer software for accessing, browsing, sharing, defining, maintaining, virtualizing and communicating information over computer networks, servers and secure private networks; computer software for use in creating, managing, controlling, changing, replicating, deploying, naming and linking information over computer networks, servers and secure private networks; computer software for assisting users in navigating through computer networks, servers and secure private networks; computer software for running web app...*

WHITE CLAW SELTZER WORKS (2016) Owner at filing: *MARK ANTHONY INTERNATIONAL SRL.* $KL_{\text{pros}} = 9.12$, $KL_{\text{retr}} = 7.35$, $\Delta KL = +1.78$.

Goods/services text: *Hard seltzer Brewed sugar-based beer*

LIQUID DEATH (2023) Owner at filing: *Supplying Demand, Inc..* $KL_{\text{pros}} = 3.97$, $KL_{\text{retr}} = 3.80$, $\Delta KL = +0.17$.

Goods/services text: *Powdered drink mixes for making flavored water, sports drinks, and energy drinks; powders and concentrates for making flavor enhanced water and sparkling water Powdered nutritional supplement drink mixes and concentrates*

HARD MTN DEW (2021) Owner at filing: *PEPSICO, INC..* $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.96$, $\Delta KL = -0.15$.

Goods/services text: *flavored brewed malt-based alcoholic beverages in the nature of beer Alcoholic flavored brewed malt beverages, except beers*

ROCKSTAR (2002) Owner at filing: *ROCKSTAR ENERGY LLC.* $KL_{\text{pros}} = 4.74$, $KL_{\text{retr}} = 3.92$, $\Delta KL = +0.81$.

Goods/services text: *Sports drinks, namely, energy drinks*